

Variables:

- n : discrete time index (generation or week)
- f_n : the true frequency of the focal allele or lineage at time n . Hidden state we want to estimate
- f_n^{obs} : is observed frequency of the focal allele at time n
- $N_e(n)$: the effective population at time n
- $T(n)$: the generation time at time n
- $\tilde{N}_e(n)$: the scaled effective population size, $\tilde{N}_e(n) = N_e(n)T(n)$
- M_n : is the total number of sequences sampled at time n
- C_n : the measurement overdispersion param, $C_n=1$ is uniform, $C_n>1$ represent overdispersed sampling noise
- W_n : process noise (stochasticity due to genetic drift)
- V_n : measurement noise (stochasticity due to the sampling and observation process)

Process Model:

First start with Expected value: find the state f_{n+1} conditioned on the previous distribution f_n . Assume neutral Wright-Fisher model (assumes the number of individuals carrying a specific allele in the next generation follows a binomial dist.)

$$E[f_{n+1}|f_n] = f_n$$

$$Q_n = \text{Var}(f_{n+1}|f_n) = \frac{f_n(1-f_n)}{\tilde{N}_e(n)}$$

We can write the state transition as a random walk driven by process noise:

$$f_{n+1} = f_n + \underbrace{W_n}_{\text{process noise}}$$

$$W_n \sim N(0, Q_n)$$

Measurement Model:

$$E[f_n^{obs}|f_n] = f_n$$

For perfect uniform Sampling Variance would be $\frac{f_n(1-f_n)}{M_n}$

to account for testing bias and measurement overdispersion

$$R_n = \text{Var}(f_n^{obs}|f_n) = C_n \frac{f_n(1-f_n)}{M_n}$$

where our measurement equation:

$$f_n^{obs} = f_n + \underbrace{V_n}_{\text{measurement noise}}$$

$$V_n \sim N(0, R_n)$$

Kalman Filter:

Lets start with a the assumption we know the true state at f_0 and our initial error covariance $P_{0|0} = 0$ Since we know the true frequency perfectly (zero estimation error)

Phase 1 prediction:

State prediction

Since we know the true state and assume neutral wright-fisher then our prediction for next step is just our true state

$$\hat{f}_{1|0} = f_0$$

Calculate process noise variance

$$Q_0 = \frac{f_0(1-f_0)}{\tilde{N}_e(0)}$$

Covariance prediction (A priori error variance)

$$P_{1|0} = 0 + Q_0 = \frac{f_0(1-f_0)}{\tilde{N}_e(0)}$$

Phase 2 Update Step:

Calculate measurement noise variance

$$R_1 = C_1 \frac{f_0(1-f_0)}{M_1} \quad \text{if perfect sequencing } R_1 = 0$$

Calculate the innovation:

$$y_1 = f_1^{obs} - f_0$$

Calculate innovation covariance

$$S_t = Q_0 + R_t$$

(Our total uncertainty is just the variance from genetic drift plus the variance from sampling noise)

Calculate the Kalman gain

$$K_t = \frac{P}{P+R} = \frac{Q_0}{Q_0+R_t}$$

State update:

$$\hat{f}_{t+1} = f_0 + K_t (f_t^{\text{obs}} - f_0)$$

Covariance update:

$$P_{t+1} = (1-K_t)Q_0$$

Connecting our notation to the papers:

First draw attention to equations 14-16 in the model section of methods

These describe the forward algorithm:

$$14) \underbrace{N(x, M_1, V_1)}_1 \underbrace{N(x, M_2, V_2)}_2 = N(M_1, M_2, V_1 + V_2) N(x, M_{12}, V_{12})$$

1. represents the prediction in our derivation where M_1 is our model's frequency ($\hat{f}_{n|n-1}$) and V_1 is our prediction error variance ($P_{n|n-1}$)

2. represents the measurement where M_2 is sequenced frequency (f_n^{obs}) and V_2 is measurement noise variance (R_n)

$$15) M_{12}(M_1, M_2, V_1, V_2) = \frac{M_1 V_2 + M_2 V_1}{V_1 + V_2}$$

$$\Rightarrow M_1 \left(\frac{V_2}{V_1 + V_2} \right) + M_2 \left(\frac{V_1}{V_1 + V_2} \right)$$

$$\Rightarrow M_1 \left(\frac{V_1 + V_2 - V_1}{V_1 + V_2} \right) + M_2 \left(\frac{V_1}{V_1 + V_2} \right)$$

$$\Rightarrow M_1 \left(1 - \frac{V_1}{V_1 + V_2} \right) + M_2 \left(\frac{V_1}{V_1 + V_2} \right)$$

$$\Rightarrow \underbrace{M_1}_{\text{a priori prediction}} + \underbrace{\frac{V_1}{V_1 + V_2}}_{K = \frac{P}{P+R}} \underbrace{(M_2 - M_1)}_{\text{innovation } (f_n^{\text{obs}} - \hat{f}_{n|n-1})}$$

To find our covariance update we can focus on

$$16) V_{12} = \frac{1}{\frac{1}{V_1} + \frac{1}{V_2}} = \frac{V_1 V_2}{V_1 + V_2} = V_1 \left(\frac{V_2}{V_1 + V_2} \right) = V_1 \left(\frac{V_1 + V_2 - V_1}{V_1 + V_2} \right)$$

$$\Rightarrow V_{12} = V_1 \left(1 - \frac{V_1}{V_1 + V_2} \right)$$

$$\text{Sub in } V_1 = P \text{ and } \frac{V_1}{V_1 + V_2} = K$$

$$V_{12} = P(1-K) \text{ which is what we had for a posteriori error cov update}$$

Likelihood and parameter choice:

The paper then uses equation 13 and 17 to find the most likely model parameters to fit the data

$$13) L(\vec{\phi}_{0:T} | \theta_{0:T}) = \prod_{\alpha} P(\phi_{0:T}^{\alpha} | \theta_{0:T}) p(\theta_{0:T}) \text{ where } \phi_{0:T}^{\alpha} = \{\phi_0^{\alpha}, \dots, \phi_T^{\alpha}\}, \theta_{0:T} = \{\tilde{N}_e(t), \dots, \tilde{N}_e(t), c_0, \dots, c_t\}$$

Uniform Prior on Params $\leftarrow$ Assume no knowledge of prior
 $P(\theta_{0:T}) \propto 1$

13) requires us to know true frequencies at every time which we don't know so intractable problem

that's where equation 17 comes in as the prediction error decomposition

$$17) p(\phi_{0:T}^{\alpha} | \theta_{0:T}) = \prod_{i=1}^T N(\phi_i^{\alpha} | M_i, \frac{c_i}{M_i} + V_i)$$

used to tell likelihood in 17 by running Kalman filter forward and looking at how badly our model's predictions missed the actual lab data at each step

Transition probability:

transition probability (prediction step) and emission probability (update step)

In pure biological systems neither of these probs are Gaussian. They are binomial distributions, since we deal with discrete counts of individuals/viruses

cannot run Kalman filter on discrete binomial distributions, so we require continuous vars and noise independent of state.

Before this we start with classic wright-fisher model

If we have N_e individuals, variant frequency f_t , the new number of individuals carrying variant in next gen is

$$n_{t+1} \sim \text{Binomial}(N_e, f_t)$$

this is just transition prob

the emission prob is if lab sequences M_t genomes, the number for variant is (k_t) which is

$$k_t \sim \text{Binomial}(M_t, f_t)$$

This relates all the way back to our equations for f_{n+1} and f_t^{obs}

we can rewrite the equations as a draw from Gaussian by central limit thm

$$f_{n+1} \sim N(f_t, \frac{f_t(1-f_t)}{N_e}) = N(f_t, Q_t)$$

$$f_t^{\text{obs}} \sim N(f_t, c_t \frac{f_t(1-f_t)}{M_t}) = N(f_t, R_t)$$

Now authors make assumption that $f_t \ll 1$, so variant we are tracking represents a small fraction of our pop.

$$\text{So } 1 - f_t \approx 1$$

$$\Rightarrow \frac{f_t(1-f_t)}{N_e} \approx \frac{f_t}{N_e}$$

$$\Rightarrow c_t \frac{f_t(1-f_t)}{M_t} \approx c_t \frac{f_t}{M_t}$$

Problem: variance still tied to state f_t . If var dependent on state its non-linear but we want to linearize so decouple with delta func

so to get around this we take square-root of $f_t \Rightarrow \phi_t = \sqrt{f_t}$

Delta method: $\text{Var}(g(x)) \approx [g'(x)]^2 \times \text{Var}(x)$

$$M = f_t, \text{Var}(f_t) = \frac{f_t}{N_e}$$

transformation function: $g(f_t) = \sqrt{f_t}$

$$\text{So } \text{Var}(\sqrt{f_t}) \approx \left(\frac{1}{2\sqrt{f_t}}\right)^2 \times \text{Var}(f_t)$$

$$\Rightarrow \left(\frac{1}{4f_t}\right) \times \left(\frac{f_t}{N_e}\right) = \frac{1}{4N_e}$$

$$\Rightarrow c_t \frac{f_t}{M_t} \approx \frac{c_t}{4M_t}$$

$$\text{So } p(\phi_{t+1} | \phi_t) = N(\phi_t, \frac{1}{4N_e})$$

$$p(\phi_t^{\text{obs}} | \phi_t) = N(\phi_t, \frac{c_t}{4M_t})$$

General Kalman filter:

Neutral wright fisher (no selection)

$$\text{State transition: } f_{n+1} = f_n + w_n$$

$$\text{Process noise: } Q_n = \text{Var}(f_{n+1} | f_n) = \frac{f_n(1-f_n)}{N_e(n)}$$

$$\text{measurement eq: } f_n^{\text{obs}} = f_n + v_n$$

$$\text{measurement noise: } \text{Var}(f_n^{\text{obs}} | f_n) = c_n \frac{f_n(1-f_n)}{M_n}$$

Phase 1: Prediction (A priori)

$$\text{State prediction: } \hat{f}_{n+1|n} = \hat{f}_n$$

$$\text{Error covariance prediction: } P_{n+1|n} = P_{n|n} + Q_n$$

Phase 2: update (A posteriori)

$$\text{Var sampling at new time step: } R_{n+1} = c_{n+1} \frac{\hat{f}_{n+1|n}(1-\hat{f}_{n+1|n})}{M_{n+1}}$$

$$\text{Innovation: difference between what we saw and expected } y_{n+1} = f_{n+1}^{\text{obs}} - \hat{f}_{n+1|n}$$

$$\text{Innovation covariance: } S_{n+1} = P_{n+1|n} + R_{n+1}$$

$$\text{Kalman Gain: } K_{n+1} = \frac{P_{n+1|n}}{S_{n+1}} \text{ if } P_{n+1|n} \text{ is very large (very uncertain about our prediction)} \quad K \rightarrow 1$$

else trust process and noisy measurement $K=0$

Final state and Covariance:

State correction: $\hat{f}_{n+1|n+1} = \hat{f}_{n+1|n} + K_{n+1}(y_{n+1})$

Error covariance: $P_{n+1|n+1} = (1 - K_{n+1})P_{n+1|n}$

Single step error analysis: Inverse of variance ($1/\sigma^2$) is known as precision

We define error at time $n+1$ as $e_{n+1} = f_{n+1} - \hat{f}_{n+1|n+1}$

Our goal is to minimize the expected square of the error which is what $P_{n+1|n+1}$ represents

Moving from $n \rightarrow n+1$ the 'error' of uncertainty in our estimate grows and then shrinks

Before our observations, our uncertainty increases due to genetic drift

$P_{n+1|n} = \underbrace{P_{n|n}}_{\text{inherited error from previous steps}} + \underbrace{Q_n}_{\text{new uncertainty introduced by genetic drift}}$

then we reduce with Kalman gain update

$P_{n+1|n+1} = (1 - K_{n+1})P_{n+1|n}$

Sub in K so

$\Rightarrow P_{n+1|n+1} = \left(1 - \frac{P_{n+1|n}}{P_{n+1|n} + R_{n+1}}\right) P_{n+1|n} \Rightarrow \left(\frac{P_{n+1|n} + R_{n+1} - P_{n+1|n}}{P_{n+1|n} + R_{n+1}}\right) P_{n+1|n} \Rightarrow \frac{R_{n+1} P_{n+1|n}}{P_{n+1|n} + R_{n+1}}$ then flip and cancel

$\Rightarrow \frac{1}{P_{n+1|n+1}} = \frac{1}{P_{n+1|n}} + \frac{1}{R_{n+1}}$ adding two positive values so resulting precision is always higher than either individual part

posterior precision = Prior precision + Measurement precision

division of allele
2, 4, 8, 16 so on

vs

error of filter
(one-step/all)

optimal coarse graining